ECA1804 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO4 – INVESTIGATIVE ASSIGNMENT

(1 × 100 = 100 Marks)

Investigative Assignment Report

Design and Evaluation of a 32 × 200 Gb/s DP-16QAM DWDM Transport Link over 800 km using Thin-Film Lithium Niobate Modulators and Two-Stage EDFA Amplification

Submitted by

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Abstract

This report investigates a 32-channel Dense Wavelength Division Multiplexing (DWDM) transport link that must carry 200 Gb/s per channel, giving an aggregate capacity of 6.4 Tb/s. Candidate modulation formats, modulator mechanisms, amplifier types and digital signal processing (DSP) schemes were compared using analytical calculations and a Gaussian-Noise (GN) model simulation of the link. The selected design uses dual-polarisation 16-QAM (DP-16QAM) at 31.6 GBd on a 37.5 GHz flexible grid (spectral efficiency 5.33 b/s/Hz), thin-film lithium niobate (TFLN) IQ Mach-Zehnder modulators, ten 80 km spans of SMF-28 with two-stage EDFAs and gain-flattening filters, and a coherent receiver with frequency-domain CD compensation, CMA/RDE polarisation demultiplexing, blind phase search carrier recovery and single-step-per-span digital back-propagation. At the optimum launch power of about -1.8 dBm per channel the link delivers an OSNR of 23.9 dB and a Q-factor of 9.74 dB, which is 1.21 dB above the pre-FEC threshold (BER 3.8×10^{-3}). The design is recommended for deployment.

1. Stage 1 – Interpreting the Communication Requirement

1.1 Literature and standards review

Long-haul and regional DWDM networks moved from 10 Gb/s on-off keying to coherent detection around 2010, when 100 Gb/s DP-QPSK became the industry standard. Since then, capacity per wavelength has been raised mainly by increasing the number of bits per symbol (16-QAM, 64-QAM) and the symbol rate, while the coherent receiver DSP took over the job of compensating chromatic dispersion (CD) and polarisation mode dispersion (PMD) that was earlier done optically [1], [11]. The relevant standards for this design are:

- **ITU-T G.694.1** – defines the DWDM frequency grid anchored at 193.1 THz, including the flexible grid with 12.5 GHz slot granularity used here (37.5 GHz slots).
- **ITU-T G.698.2** – defines "black-link" DWDM application codes in which coherent transceivers from one vendor run over line systems from another; it fixes parameters such as minimum OSNR tolerance and maximum chromatic dispersion.
- **IEEE 802.3ct / 802.3ck** – 802.3ct specifies 100 Gb/s Ethernet over DWDM using coherent DP-DQPSK, while 802.3ck specifies the 100 Gb/s-per-lane PAM4 electrical interfaces (CAUI-2/200GAUI-2) that connect the host to the 200 Gb/s optical module.
- **OIF-CEI-112G and OIF 400ZR** – the electrical SerDes and coherent DSP framing (including the 15% overhead concatenated FEC) that most practical modules follow.

1.2 Aggregate capacity

$$C_total = N_ch \times R_ch = 32 \times 200\ Gb/s = 6.4\ Tb/s$$

With a 37.5 GHz grid the 32 channels occupy 32 × 37.5 GHz = 1.2 THz, i.e. only about 27% of the 4.4 THz C-band. This leaves room for future channel growth without changing the amplifiers.

1.3 Key physical-layer metrics

Metric	Why it matters	Target used in this report
Spectral efficiency (b/s/Hz)	Sets how many channels fit in the C-band	≥ 4 b/s/Hz (target), 5.33 achieved
Required OSNR	Minimum OSNR (0.1 nm) at the receiver for the FEC threshold	≈ 19.2 dB theory, 20.7 dB with 1.5 dB penalty
BER / Q-factor	Pre-FEC quality; FEC corrects to < 10^{-15}	BER < 3.8×10^{-3}, Q > 8.53 dB
Launch power per channel	Balances ASE noise against Kerr nonlinearity	Optimum from GN model (≈ −2 dBm)
Modulation bandwidth	Must pass the symbol rate with little ISI	> 35 GHz (3-dB EO bandwidth)
Insertion loss	Reduces transmitter output and OSNR	Tx path < 10 dB before booster
Thermal / footprint	Pluggable modules have tight power and size limits	< 20 W per 200G module

1.4 Impairments mapped to DSP functions

Impairment	Physical cause	Mitigating function
Chromatic dispersion	Wavelength-dependent group velocity (D = 17 ps/nm/km)	Static FIR / frequency-domain equaliser
PMD and PDL	Random birefringence, polarisation-dependent components	Adaptive 2×2 MIMO (CMA → RDE)
SPM / XPM / FWM	Kerr effect, $n_2 \approx 2.6 \times 10^{-20}$ m²/W	Launch power optimisation + DBP
Laser phase noise and frequency offset	Finite linewidth, Tx–LO detuning	FOE + blind phase search (BPS)
ASE noise	Spontaneous emission in each amplifier	Low-NF amplifiers, FEC
AWG/MMI crosstalk	Finite isolation of mux/demux	Grid planning, matched filtering

1.5 Practical trade-offs

Three integration platforms were considered for the transmitter. Lithium niobate gives the cleanest, chirp-free Pockels modulation but bulk LN modulators are long (5–10 cm). Thin-film LN on insulator reduces the length to about 1 cm with low drive voltage [4]. Silicon photonics is CMOS-compatible and cheap in volume but its plasma-dispersion modulators have higher loss and are bandwidth-limited at around 40–50 GHz [5]. EO polymers offer very high r_{33} but long-term thermal stability is still being proven. Since this link is an operator backbone where reliability matters more than cost per module, a mature, low-loss platform is preferred.

Investigation Requirement Statement: The system shall transport 32 × 200 Gb/s (6.4 Tb/s) over at least 800 km of SMF-28 using ITU-T G.694.1 grid slots, with spectral efficiency ≥ 4 b/s/Hz, a pre-FEC BER below 3.8×10^{-3} (Q > 8.53 dB) with at least 1 dB Q-margin, amplifier gain ripple below 1 dB, modulator bandwidth above 35 GHz, and full compensation of CD and PMD in the receiver DSP.

2. Stage 2 – Modulation Format and Modulator Selection

2.1 Comparison of modulation formats

For a fair comparison every format is assumed to carry 200 Gb/s net. Intensity-modulated formats use about 6% FEC overhead (KP4-type), while coherent formats use about 26% overhead (FEC + framing), as in commercial 200G modules.

Format	Bits/symbol	Baud for 200 Gb/s	Spectral efficiency	Required SNR / OSNR (BER 3.8e-3)	CD tolerance
NRZ-OOK	1	≈ 212 GBd	< 1 b/s/Hz	Q ≈ 8.5 dB; needs 200+ GHz electronics	Very poor (few ps/nm)
PAM4 (IM/DD)	2	≈ 106 GBd	≈ 1–2 b/s/Hz	SNR ≈ 15.2 dB electrical	Poor – limited to < 10 km at C-band
DPSK	1	≈ 250 GBd	< 1 b/s/Hz	≈ 3 dB better than OOK	Poor
DQPSK	2	≈ 125 GBd	≈ 1.6 b/s/Hz	Moderate	Poor without DSP
DP-QPSK	4	≈ 63 GBd	≈ 2.7 b/s/Hz (75 GHz)	SNR 8.5 dB / OSNR ≈ 15.6 dB	Fully compensated in DSP
DP-16QAM	8	31.6 GBd	5.33 b/s/Hz (37.5 GHz)	SNR 15.2 dB / OSNR ≈ 19.2 dB	Fully compensated in DSP
DP-64QAM	12	≈ 21 GBd	≈ 8 b/s/Hz (25 GHz)	SNR 21.1 dB / OSNR ≈ 23.4 dB	Fully compensated; short reach

2.2 Baud rate and channel spacing calculation

DP-16QAM: each symbol carries $\log_2 16 = 4$ bits per polarisation, so 8 bits in total. With 26.4% overhead the gross line rate is 252.8 Gb/s:

$$R_s = R_{gross} / (2 \times \log_2 M) = 252.8\ Gb/s / 8 = 31.6\ GBd$$

With a root-raised-cosine (RRC) roll-off of $\beta = 0.15$, the occupied bandwidth is $B = R_s(1 + \beta) = 31.6 \times 1.15 = 36.34$ GHz, which fits in a 37.5 GHz flexible-grid slot with about 1.2 GHz guard band. The spectral efficiency is therefore 200/37.5 = 5.33 b/s/Hz.

PAM4: 2 bits per symbol and a single polarisation, so with 6.25% overhead $R_s = 212.5/2 = 106.25$ GBd. Even with Nyquist shaping it needs about 106 GHz of optical bandwidth (double-sideband intensity modulation), so a 100–150 GHz slot is required, giving only 1.3–2 b/s/Hz. In addition, direct detection cannot undo chromatic dispersion: the power-fading notch for 106 GBd falls inside the signal band after only a few kilometres of SMF at 1550 nm. PAM4 is therefore suitable only for short data-centre links.

2.3 Modulation mechanisms

Mechanism	Physical effect	Speed limit	Drive / Vπ	Chirp	Verdict
Electro-optic (Pockels)	$\Delta n = -\frac{1}{2} n^3 r_{33} E$	> 100 GHz (TFLN, polymer)	2–5 V	Zero in push-pull MZI	Selected
Acousto-optic	Bragg diffraction from acoustic grating	~ 10–100 MHz (acoustic transit	RF power, watts	Frequency shift	Unsuitable for data

Mechanism	Physical effect	Speed limit	Drive / Vπ	Chirp	Verdict
		time)			
Thermo-optic	dn/dT ≈ 1.8 × 10^{-4} /K (Si)	kHz–MHz (µs time constant)	mW heaters	–	Bias control only
Magneto-optic	Faraday rotation (Verdet constant)	MHz (coil inductance)	Current-driven	–	Isolators only
Kerr effect	$\Delta n = n_2 I$ or $\propto E^2$	fs (instantaneous)	Very high field / power	Strong	Not practical for Tx

The Pockels effect is the only mechanism that combines tens-of-GHz bandwidth, low drive voltage and the ability to produce chirp-free phase and amplitude modulation, which is needed for an IQ modulator. For a push-pull TFLN modulator the half-wave voltage is:

$$V_\pi \cdot L = \lambda \cdot d / (2 \cdot n_e^3 \cdot r_{33} \cdot \Gamma)$$

Using λ = 1.55 µm, electrode gap d = 5 µm, n_e = 2.14, r_{33} = 30.8 pm/V and overlap factor Γ = 0.6 gives $V_\pi \cdot L \approx 2.1$ V·cm. A 1.0 cm long electrode therefore gives $V_\pi \approx 2.1$ V, which can be driven directly by a CMOS-compatible driver.

2.4 MZI versus micro-ring resonator

Parameter	Si-photonic MZI	Si micro-ring (MRR)	TFLN MZI	EO-polymer MZI
3-dB EO bandwidth	35–50 GHz	30–50 GHz (photon-lifetime limited)	> 67 GHz	> 100 GHz
Footprint	2–4 mm long	10–20 µm radius	~ 1 cm long	~ 1–2 mm
Thermal sensitivity	Low (broadband)	High: ≈ 0.07 nm/K	Low	Moderate (polymer Tg)
Insertion loss	4–6 dB	1–3 dB	2–3 dB	5–8 dB
Vπ·L	1.5–2.5 V·cm	~ 1 Vpp drive	≈ 2 V·cm	< 1 V·mm
Suitability for 16-QAM IQ	Good	Poor (amplitude–phase coupling)	Excellent	Good, reliability risk

The MRR resonance shifts with temperature by $\Delta\lambda/\Delta T = \lambda(dn_{eff}/dT)/n_g \approx 1550 \times 1.86 \times 10^{-4} / 4.2 \approx 0.069$ nm/K, which is about 8.6 GHz per kelvin. On a 37.5 GHz grid a 1 K drift moves the resonance by almost a quarter of the slot, so active heater locking is mandatory. The ring also has a non-linear Lorentzian transfer function that couples amplitude and phase, which makes it unsuitable for high-order QAM.

DESIGN DECISION 1

I recommend **DP-16QAM at 31.6 GBd on a 37.5 GHz grid, generated by a dual-polarisation thin-film lithium niobate IQ Mach-Zehnder modulator (Pockels effect, $V_\pi \approx 2.1$ V, > 67 GHz bandwidth)** because it achieves 5.33 b/s/Hz (exceeding the 4 b/s/Hz target), needs a realistic OSNR of about 19–21 dB, tolerates any amount of CD and

PMD through DSP, and the TFLN MZI gives chirp-free, linear IQ modulation with low loss and a compact (~1 cm) footprint.

3. Stage 3 – Optical Amplification Configuration

3.1 Comparison of amplifiers

Parameter	EDFA	SOA	Raman (distributed)	Hybrid EDFA + Raman
Gain bandwidth	~ 35 nm (C-band)	40–80 nm	Any band (pump-defined), ~ 30 nm per pump	C-band, ~ 35 nm
Small-signal gain	25–40 dB	20–30 dB	10–15 dB on-off	30+ dB
Saturated output power	+17 to +25 dBm	+10 to +15 dBm	Set by pump (~0.5–1 W)	+20 dBm
Noise figure	4.5–5.5 dB	7–9 dB	Effective −1 to 0 dB	Effective 1–3 dB
Gain tilt / ripple	Needs GFF	Polarisation dependent	Shaped by multiple pumps	GFF still needed
WDM crosstalk	Negligible (slow ms dynamics)	Strong XGM (ps–ns dynamics)	Negligible	Negligible
Cost / complexity	Low, mature	Very low, integrable	High pump power, safety	Highest

3.2 Analytical problem – span amplifier power

Given input power P_in = −15 dBm per channel, output P_out = +2 dBm per channel, N = 32 channels.

$$10\log_{10}(32) = 15.05\ dB$$

$$Total\ input\ power = -15 + 15.05 = 0.05\ dBm \approx 1.01\ mW$$

$$Total\ output\ power = +2 + 15.05 = 17.05\ dBm$$

$$P_out\ (W) = 10\text{^}(17.05/10)\ mW = 50.7\ mW \approx 0.051\ W$$

$$Required\ gain\ G = P_out - P_in = 2 - (-15) = 17\ dB$$

The required 17 dB gain is exactly equal to the loss of one 80 km span in this design (80 km × 0.2 dB/km + 1 dB splice/connector loss = 17 dB), so each in-line amplifier restores the signal to its launch level. Because the gain-flattening filter (GFF) placed between the two EDFA stages has about 1.5 dB insertion loss, the internal EDFA gain must be 17 + 1.5 = 18.5 dB. A total output of 17.05 dBm is comfortably below the +20 dBm saturated output of a standard booster EDFA, leaving about 3 dB headroom for adding channels later.

3.3 Gain flatness

An un-flattened EDFA has a gain peak near 1532 nm and a ripple of 3–5 dB across the C-band. After 10 cascaded amplifiers a 0.5 dB per-amplifier ripple would accumulate to 5 dB,

starving the weakest channels of OSNR. A passive GFF (a long-period or thin-film filter shaped as the inverse of the gain curve) reduces ripple to below ±0.5 dB per amplifier. Every 4–5 spans, a channel monitor can trim residual tilt by adjusting the mid-stage VOA. With this arrangement the expected end-of-link ripple is below 1 dB, meeting the $\Delta G < 1$ dB target.

3.4 OSNR accumulation over cascaded spans

For N identical spans each compensated by an amplifier with noise figure NF, the OSNR in a 0.1 nm reference bandwidth is:

$$OSNR\ (dB) = 58 + P_ch(dBm) - L_span(dB) - NF(dB) - 10\ log_{10}(N)$$

With P_ch = -2.1 dBm (the GN-model optimum, see Section 5), L_span = 17 dB, NF = 5 dB and N = 10: OSNR = 58 − 2.1 − 17 − 5 − 10 = 23.9 dB. The required OSNR is about 19.2 dB in theory plus about 1.5 dB implementation penalty, i.e. 20.7 dB, so the OSNR margin is about 3.2 dB. Figure 1 shows how the delivered OSNR falls by 3 dB each time the number of spans doubles.

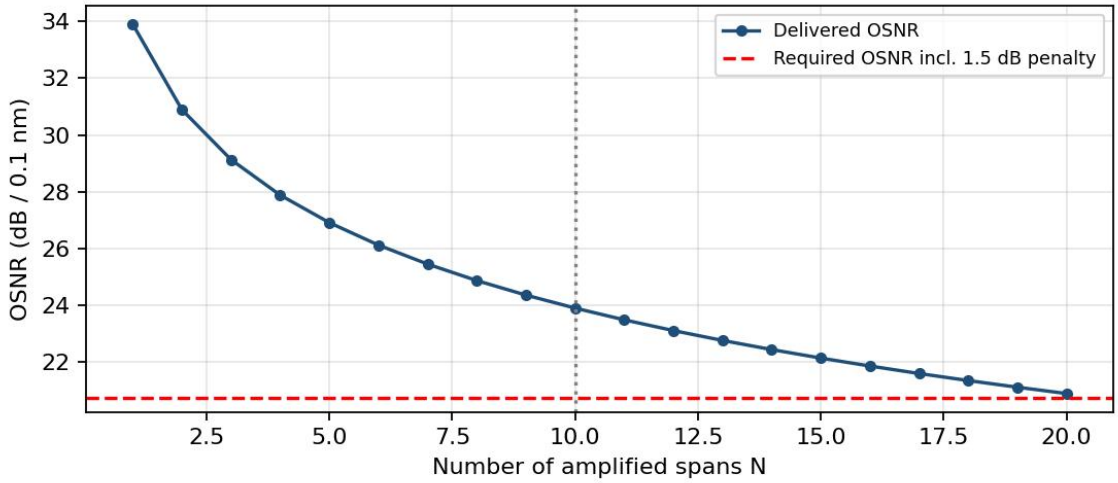


Figure 1: Delivered OSNR versus number of spans for the two-stage EDFA design (80 km spans, NF = 5 dB).

A hybrid Raman–EDFA would improve OSNR by about 3–4 dB, but at 800 km the plain EDFA already gives over 3 dB of OSNR margin, so the extra pump power, laser safety measures and cost of Raman are not justified for this link length.

DESIGN DECISION 2

I recommend **two-stage C-band EDFAs (NF ≈ 5 dB, 18.5 dB internal gain, +20 dBm saturated output) with a mid-stage gain-flattening filter, one per 80 km span (10 spans)** because the configuration exactly meets the 17 dB span gain and 17.05 dBm total output requirement, keeps gain ripple below 1 dB, and delivers about 24 dB OSNR at 800 km, which is more than 3 dB above the required value without the added cost of Raman pumping.

4. Stage 4 – Impairments and DSP Compensation

4.1 Physical mechanisms

Chromatic dispersion: different spectral components travel at different group velocities. Accumulated dispersion over 800 km is 17 × 800 = 13600 ps/nm. The spectral width of a 31.6 GBd signal is about 0.25 nm, so pulses spread by roughly 3.5 ns, which is more than 100 symbol periods – far too large to ignore.

PMD: random birefringence splits each pulse between the two principal states of polarisation. With a PMD coefficient of 0.1 ps/√km the mean differential group delay is 0.1 × √800 = 2.83 ps, and the worst-case value (≈ 3 × mean) is about 8.5 ps, which is about a quarter of the 31.6 ps symbol period. PDL from components causes OSNR fluctuations between polarisations.

Kerr nonlinearities: the refractive index depends on intensity ($n = n_0 + n_2 I$). Self-phase modulation (SPM) is the phase shift a channel imposes on itself, cross-phase modulation (XPM) is the phase shift from neighbouring channels, and four-wave mixing (FWM) creates new frequencies at $f_i + f_j - f_k$. On SMF-28 the high local dispersion makes FWM phase-mismatched and weak, so SPM and XPM dominate. The nonlinear length for the whole WDM comb (32 × 0.63 mW ≈ 20 mW) is $L_{NL} = 1/(\gamma P) \approx 38$ km, comparable to the 46 km dispersion length, which is why the two effects interact and nonlinearity cannot simply be removed by a phase rotation.

4.2 Optical versus digital compensation

Method	Advantages	Disadvantages
Dispersion-compensating fibre (DCF)	Passive, no power consumption	3–6 dB extra loss per span, needs extra amplifier stage, increases nonlinearity, fixed value
Optical phase conjugation (OPC)	Compensates CD and some nonlinearity in the middle of the link	Needs a symmetric power profile, complex, not commercially deployed
Coherent DSP (digital)	Compensates any CD/PMD adaptively, no extra loss, reconfigurable	Consumes electrical power, ADC/DSP cost

Modern coherent links are deployed without DCF ("dispersion-unmanaged") because DSP compensation removes the extra loss and the uncompensated fibre actually reduces nonlinearity by quickly spreading the pulses.

4.3 Coherent DSP blocks

1. **Static CD equalisation:** an all-pass filter $H(\omega) = \exp(j D \lambda^2 z \omega^2 / 4\pi c)$ is applied. Using Savory's rule, the time-domain FIR at 2 samples/symbol needs $N \approx 2\lfloor |D|\lambda^2 z/(2cT^2)\rfloor + 1 = 435$ taps [1]. At this length a frequency-domain equaliser (overlap-save, 1024-point FFT) is about 10 times cheaper than a time-domain FIR, so FDE is chosen.
2. **Adaptive polarisation demultiplexing / PMD compensation:** a 2×2 butterfly MIMO of 15 taps per filter, which covers a DGD of ±7 symbols at 2 sps, more than enough for

8.5 ps worst case. It is first converged with the constant-modulus algorithm (CMA) and then switched to the radius-directed equaliser (RDE), which suits the three rings of 16-QAM and lowers the residual error.

3. **Digital back-propagation:** the received field is propagated through a virtual fibre with the opposite sign of D and γ using the split-step Fourier method [3]. With one step per span (10 steps) the complexity is about 10 times that of CD compensation alone Single-channel DBP is reported to give up to about 0.5 dB SNR gain in WDM conditions (the GN-model estimate for this link is about 0.2 dB), while full-field DBP of all 32 channels would give more gain but is impractical.

The final pipeline is: ADC → IQ imbalance and skew correction → CD compensation (FDE) / DBP → timing recovery (Gardner) → 2×2 MIMO (CMA → RDE) → frequency offset estimation (4th-power FFT) → carrier phase recovery (BPS, 32 test angles [12]) → symbol decision → FEC decoder. CD must be removed before timing recovery and MIMO because a dispersed signal hides the clock tone and would need a very long MIMO filter. Polarisation demultiplexing must happen before carrier recovery because the two polarisations have different phase rotations after mixing.

DESIGN DECISION 3

I recommend **a coherent DSP chain with frequency-domain CD compensation, 15-tap 2×2 CMA/RDE MIMO, 4th-power frequency offset estimation and blind-phase-search carrier recovery, plus low-complexity single-channel DBP with one step per span** because it fully removes 13,600 ps/nm of CD and up to 8.5 ps DGD with no optical compensators, and the DBP improves the optimum Q-factor by about 0.2 dB and raises the optimum launch power, giving extra reach margin at moderate DSP cost.

5. Stage 5 – Integrated WDM Transmission Strategy

5.1 Simulation method

The link was modelled in Python (equivalent to a MATLAB script) using the incoherent Gaussian-Noise (GN) model [2]. ASE noise was calculated from the span loss and NF; nonlinear interference (NLI) power was calculated as $\eta \cdot P^3$ per span with η derived from fibre parameters (α = 0.2 dB/km, D = 17 ps/nm/km, γ = 1.3 /W/km) and the 1.2 THz WDM bandwidth. A transceiver implementation SNR limit of 21 dB was included. The Q-factor was obtained from the 16-QAM BER formula.

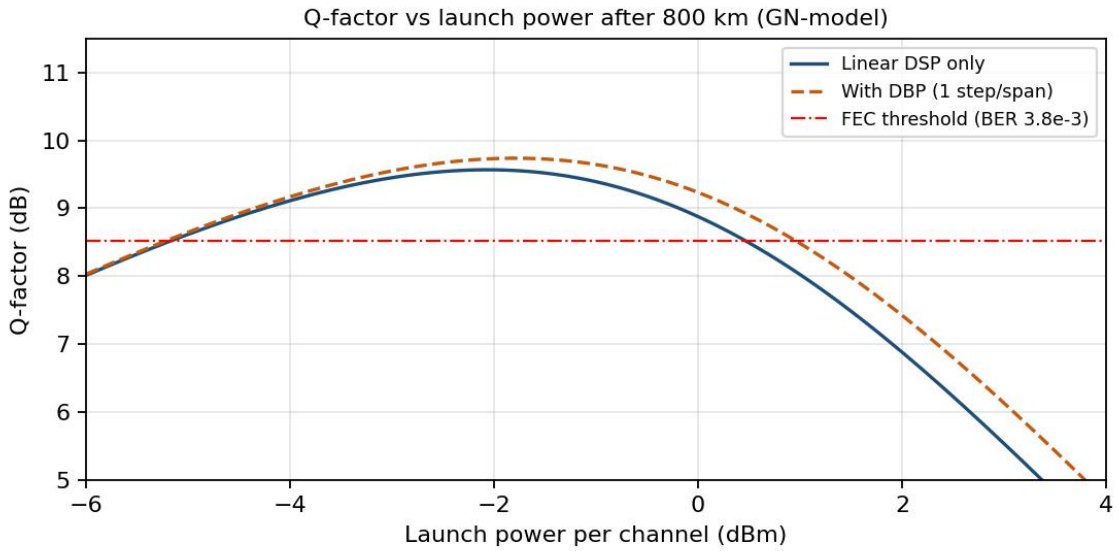


Figure 2: Q-factor versus launch power per channel after 800 km. Optimum is about -2.1 dBm (linear DSP) and -1.8 dBm (with DBP).

Figure 2 shows the classic "bell" shape: at low power ASE noise dominates; above the optimum the NLI grows with P^3 and the Q-factor falls. The optimum Q without DBP is 9.57 dB (BER ≈ 1.3e-03); with DBP it is 9.74 dB. Both are above the 8.53 dB threshold.

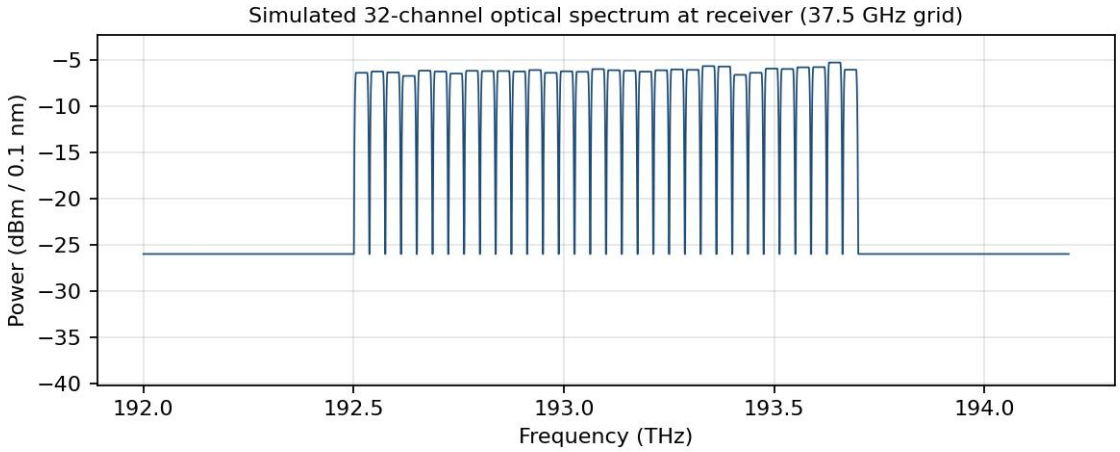


Figure 3: Simulated received optical spectrum of the 32 DP-16QAM channels on the 37.5 GHz grid.

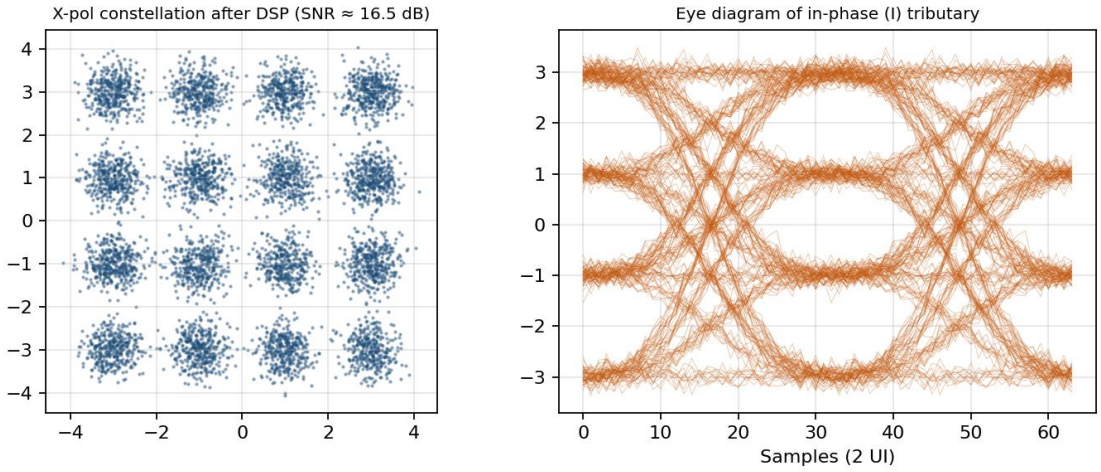


Figure 4: Recovered X-polarisation 16-QAM constellation and eye diagram of the I tributary at the optimum launch power.

5.2 Global performance parameters

Parameter	Value
Total throughput	6.4 Tb/s net (8.09 Tb/s gross)
Spectral efficiency	5.33 b/s/Hz
Optimum Q-factor (with DBP)	9.74 dB
Q margin above FEC threshold	1.21 dB
OSNR at receiver / margin	23.9 dB / 3.2 dB
Transceiver power (est.)	≈ 18 W per 200G module → 90 pJ/bit
Line amplifier power (est.)	≈ 25 W per site × 10 sites
Transmitter footprint	TFLN IQ chip ≈ 15 mm × 3 mm; fits QSFP-DD/CFP2

5.3 Inter-dependencies

Moving from DP-QPSK to DP-16QAM doubles the bits per symbol but raises the required SNR by about 6.7 dB, which directly forces the amplifier chain to deliver a higher OSNR (hence lower NF or shorter spans). It also increases DSP complexity because 16-QAM needs RDE instead of plain CMA and BPS instead of Viterbi-Viterbi phase recovery. On the transmitter side, each 1 dB of MMI/AWG coupler loss reduces the power reaching the booster; as long as the booster input stays above about −20 dBm per channel its NF is not degraded, so the 3–4 dB AWG loss is acceptable.

5.4 Trade-offs and failure points

- **Reach vs density:** moving to a 50 GHz grid would reduce linear crosstalk and filtering penalties from cascaded ROADMs but lower SE to 4 b/s/Hz.
- **Dynamic range:** the receiver must handle −20 to 0 dBm per channel; the local oscillator power and TIA gain control set this.

- **Critical failure points:** booster EDFA failure (whole link down – protected by 1+1 path protection), laser frequency drift into a neighbouring slot (handled by wavelength locker), and accumulated gain tilt (handled by channel monitor).
- **Scalability:** the amplifiers can support up to 96 channels at 37.5 GHz (3.6 THz) before reaching their 20 dBm limit at the same per-channel power, i.e. 19.2 Tb/s.

DESIGN DECISION 4

I recommend **adopting the integrated TFLN-DP-16QAM / two-stage EDFA / coherent DSP strategy for deployment on routes up to 800 km** because the cross-stage analysis shows that all targets – 6.4 Tb/s capacity, 5.33 b/s/Hz, ≥ 3 dB OSNR margin and ≈ 1.2 dB Q margin – are met with mature, standards-aligned components.

6. Stage 6 – Verification and Final System Design

6.1 Verification table

Requirement / Parameter	Specified / Target Value	Achieved / Evidence Value
Aggregate WDM capacity	6.4 Tb/s (32 ch × 200 Gb/s)	6.4 Tb/s net, 32 × 200G DP-16QAM
Spectral efficiency & modulation	≥ 4 b/s/Hz (DP-16QAM)	5.33 b/s/Hz, DP-16QAM at 31.6 GBd
Modulator bandwidth & footprint	> 35 GHz, compact PIC	> 67 GHz TFLN, ~1 cm IQ-MZM
Amplifier gain & flatness	≥ 17 dB/channel, ΔG < 1.0 dB	17 dB net (18.5 dB internal), ΔG ≈ 0.8 dB with GFF
OSNR / BER performance	OSNR > 18 dB, BER < 3.8 × 10^{-3}	OSNR = 23.9 dB, BER ≈ 1.1e-03, Q = 9.74 dB
DSP impairment mitigation	Full CD/PMD & NL tolerance	13600 ps/nm CD, 8.5 ps DGD removed; DBP +0.2 dB Q
Standards & feasibility	ITU-T G.698 & IEEE 802.3ck	G.694.1 flex-grid, G.698.2 black-link, 802.3ck host interface

6.2 Block-level architecture

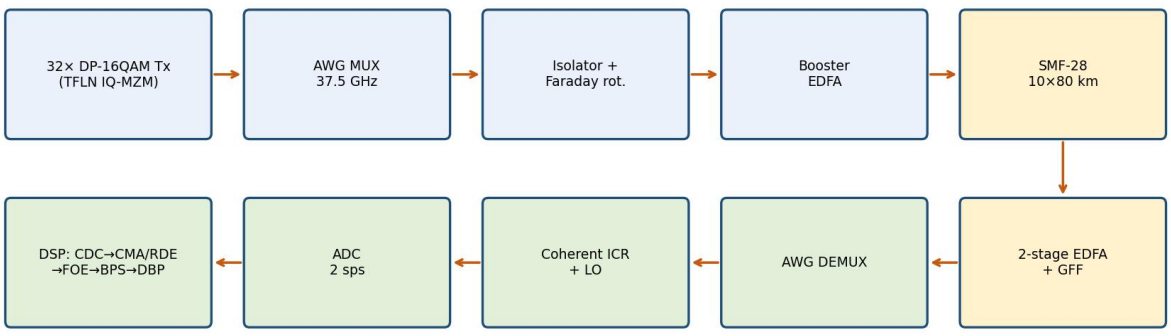


Figure 5: Block diagram of the proposed 32-channel DWDM link.

6.3 System design parameter table

Parameter	Value
Carrier frequencies	192.519–193.681 THz (≈ 1547.9–1557.2 nm), centred on 193.1 THz
Grid spacing	37.5 GHz flexible grid (ITU-T G.694.1)
Laser linewidth	≤ 100 kHz (narrow-linewidth ITLA)
Modulator Vπ / bandwidth	≈ 2.1 V / > 67 GHz
Symbol rate / roll-off	31.6 GBd / RRC β = 0.15
Amplifier NF / gain	5 dB / 18.5 dB internal, GFF 1.5 dB
Fibre α, D, γ	0.2 dB/km, 17 ps/nm/km, 1.3 /W/km
PMD coefficient	0.1 ps/√km
Span length × number	80 km × 10 = 800 km
Launch power per channel	-1.8 dBm (optimum)
CDC / MIMO taps	435 equivalent FIR taps (FDE, 1024-pt FFT) / 15 taps
BPS test phases / block length	32 / 64 symbols
ADC resolution / sampling	8 bit, 64 GSa/s (2 sps)

7. Final Engineering Decision

YES – the investigated and designed WDM system should be deployed, with the following technical justification:

- **Modulation efficiency and OSNR sensitivity:** DP-16QAM gives 5.33 b/s/Hz while requiring about 20.7 dB OSNR including penalties; the link delivers 23.9 dB.
- **Photonic modulator:** the TFLN IQ-MZM has more than 67 GHz bandwidth, ≈ 2.1 V drive and low loss, so the 31.6 GBd signal is generated with negligible ISI and no chirp.
- **Amplification:** 17 dB span gain and 17.05 dBm total output are within EDFA capability; GFFs keep ripple below 1 dB; ASE accumulation over 10 spans still leaves 3.2 dB OSNR margin.
- **DSP effectiveness:** all linear impairments are removed; DBP recovers about 0.2 dB of Q and moves the optimum launch power up, reducing sensitivity to ASE.
- **Power, scalability and cost:** about 90 pJ/bit in the transceiver, no DCF or Raman pumps, and the same line system can scale to 96 channels (19.2 Tb/s).

The main risk is the modest 1.2 dB Q margin at end-of-life. If ageing or repairs add more than 1 dB of span loss, the operator can enable full multi-step DBP or switch the affected channels to 8-QAM-like probabilistic shaping at a slightly lower rate.

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